ExoMiner Pipeline Outputs

For Pipeline Version 2.1

Miguel J. S. Martinho [*1,2], **Hamed Valizadegan** [1,2], **Jon M. Jenkins** [2], **Douglas A. Caldwell** [3,2], **Ben Tofflemire** [3,2], **and Marziye Jafariyazani** [3,2]

[1] ***KBR, Inc., Mountain View, CA 94043, USA***
[2] ***NASA Ames Research Center (NASA ARC), Moffett Field, CA 94035, USA***
[3] ***The SETI Institute, Mountain View, CA 94043, USA***

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[*] Email: miguel.martinho@nasa.gov

1 Introduction

The ExoMiner Pipeline was designed as an end-to-end pipeline to aid expert vetters in the vetting and validation of planet candidates from the TESS Mission. For a comprehensive documentation on downloading, installing, and using this pipeline, please refer to the documentation guide within the project's official NASA GitHub repository [NAS26a]. This document is intended to describe the outputs of this pipeline for anyone interested in using the pipeline results towards exoplanet candidate vetting.

The `ExoMiner Pipeline` generates two main products: a table containing the prediction scores for the SPOC TCEs detected in the set of queried TIC IDs and corresponding sector runs; and a summary report that can be optionally generated for each TCE, and includes relevant diagnostic plots that show the preprocessed data that are ingested by the model to produce the aforementioned prediction scores. In the following sections we describe these two products in detail.

2 Prediction Table

The prediction table is the final product generated by the `ExoMiner Pipeline`. It is exported as a comma-separated values (CSV) file containing two main sections: a metadata header and the tabular prediction data.

Metadata Header

The file begins with a metadata header, denoted by lines starting with the `#` character. This line preserves run provenance and configuration details. Key information captured in this header includes:

- **Pipeline Versioning:** Git commit hash, release date, and version number (e.g., Version 2.1).
- **Run Configuration:** name of the pipeline run, date the run started, classification task (e.g., `phot-vetting`), type of ExoMiner model used (e.g., single or ensemble model variant), data collection mode (either 2min or FFI), and the data sources utilized for stellar parameters and Renormalized Unit Weight Error (RUWE).
- **Classification Threshold & Label Map:** Definitions for the target classes; these include Planet Candidate (PC), Astrophysical False Positive (AFP), and Non-Transiting Phenomena (NTP) for photometric vetting, and Planet and Non-Planet for the planet validation task. For the 3-class photometric vetting task, each TCE is assigned the disposition of the class that scored the highest; for the 2-class planet vs non-planet task, the classification threshold defines the decision threshold for assigning a planet/non-planet disposition.

Data Columns

Following the metadata header, the file provides the model's predictions for each evaluated TCE. The tabular data columns are defined in Table 1.

Example Output

Figure 1 provides an example of the raw CSV output generated by the `ExoMiner Pipeline`, showing the header and two evaluated TCEs.

3 Summary Report

As supporting products, the pipeline can generate a summary report in PDF format for each SPOC TCE detected for the queried TIC IDs. These reports are optional products generated by the `ExoMiner Pipeline` and are intended to provide:

- Outputs generated by the data preprocessing pipeline to visualize the input features (e.g., flux time series, difference images) ingested by the model, derive insights about model behavior, and identify opportunities to improve both the data preprocessing pipeline and model architecture for better performance.
- Plots and visualizations similar to those used by expert vetters for vetting of planet candidates. The Data Validation reports generated by the SPOC pipeline [Jen+16] were used as a guideline when designing the summary report provided by the `ExoMiner Pipeline`, although both contain plots that are not shared.

In the following sections, we describe each plot and image shown in the summary report using 2-min SPOC TCE `TIC 347538174.1` in S14-S86 as an example.

3.1 Flux and Centroid Offset

The phase-folded and binned flux and centroid plots, provided in Figure 4, are a condensed representation of most of the time series data provided as input to the model. The top of the page includes the SPOC TCE's ephemerides along with several TCE fit parameters such as transit depth, MES, and planet radius. These values are sourced from the SPOC DV results. The stellar parameters provided by the source catalog (e.g., TICv8) are also shown. The names in the header follow the nomenclature used in the SPOC pipeline and can be found in the Exoplanet Archive Kepler's TCEs data columns definitions table [NAS26b].

The plots are generated by detrending the corresponding time series (i.e., flux, flux-weighted centroid motion) using a Savizkty-Golay filter and phase-folding using the epoch and orbital period of the detected SPOC TCE. Each time series datum (i.e., black dot) represents the median value within the bin, while

Table 1: Description of the data columns provided in the ExoMiner Pipeline prediction table output.

Column Name	Description
uid	Unique SPOC TCE identifier formatted as <tic_id>-<tce_plnt_num>-S<sector_run_id> (e.g., 1942969-1-S14-86).
target_id	TIC ID of the target star (integer scalar).
sector_run	SPOC sector run identifier. Represented as a single sector or a multi-sector range (e.g., 14-86).
tce_plnt_num	SPOC TCE planet number (integer scalar).
tce_period	TCE period in days (float scalar).
tce_duration	TCE transit duration in hours (float scalar).
tce_time0bk	TCE time of first transit in BTJD (float scalar).
tce_depth	TCE transit depth in parts per million (ppm).
ruwe	RUWE from Gaia (float scalar).
tce_prad	TCE planet radius in Earth radii (float scalar).
tce_max_mult_ev	TCE maximum multiple event statistic (float scalar).
matchedToiId	Matched TOI ID from the SPOC Data Validation (DV) report (e.g., 5459.01), or not_matched if no match is found.
score_<disp>	The ExoMiner mean score for the respective disposition class calculated over the models in the ensemble (when using a single model, this is simply the score output by the model). For the photometric vetting model, these three class probabilities sum to 1. For the Planet vs. Non-planet validation model, there is a single score for the Planet class; the score for the Non-Planet class is given by one minus the Planet score.
score_std_<disp>	The ExoMiner standard deviation score for each class calculated over the models in the ensemble.
prediction	The final predicted disposition assigned by the pipeline, based on the classification threshold and the model score.

```
# Run: exominer-pipeline-run_tess-spoc-2min_s14-s86_test-deliver-poc_6-15-2026_0833
# Created: 2026-06-15T20:15:37.934131
# Version Number: 2.1
# Git Commit Hash: 9ee77fde310cef20b470888ee5392a2887b77fc4
# Release Date: 2026-06-29 07:33:41+00:00
# Task: phot-vetting
# ExoMiner Model: full_cv_ensemble
# Label Map: {'PC': 1, 'AFP': 2, 'NTP': 0}
# Data Collection Mode: 2min
# Stellar Parameters Source: ticv8
# RUWE Source: gaiadr2
# Classification Threshold: 0
# Disposition Definitions: ========================
# PC: planet candidate
# AFP: astrophysical false positive
# NTP: non-transiting phenomena
# Column Definitions: ========================
# Column: uid: Unique SPOC TCE identifier with pattern <tic_id>-<tce_plnt_num>-S<sector_run_id>
    (e.g., 123456789-1-S1, 123456789-1-S1-3) (string)
# Column: target_id: TIC ID of the target star ( int)
# Column: sector_run: SPOC sector run identifier. If single sector run follows format <sector>.
    If multisector runs follows format <start_sector>-<end_sector> (string)
# Column: tce_plnt_num: SPOC TCE planet number (int)
# Column: tce_period: TCE period in days (float)
# Column: tce_duration: TCE duration in hours (float)
# Column: tce_time0bk: TCE time of first transit in BTJD (float)
# Column: tce_depth: TCE depth in ppm (float)
# Column: ruwe: Renormalized Unit Weight Error from Gaia (float)
# Column: tce_prad: TCE planet radius in Earth radii (float)
# Column: tce_max_mult_ev: TCE max multiple event statistic (float)
# Column: matchedToiId: matched TOI ID from SPOC DV (e.g., "1234.01" or "not_matched" if no
    match found) (string)
# Column: score/score_<disp>: The ExoMiner prediction score (mean score if using an ensemble). (
    float). For Photometric Vetting, this generates three columns (score_PC, score_AFP,
    score_NTP) representing the scores for each disposition category, which sum to 1.0. For
    Planet Validation, this generates a single score column where a value closer to 1.0
    indicates higher confidence that the TCE is a true planet.
# Column: score_std/score_std_<disp>: The uncertainty in the ExoMiner prediction, calculated as
    the standard deviation across an ensemble (will be 0.0 if using a single model). (float).
    For Photometric Vetting, this generates three columns (score_std_PC, score_std_AFP,
    score_std_NTP) representing the uncertainty for each respective disposition. For Planet
    Validation, this generates a single score_std column representing the uncertainty of the
    validation score.
# Column: prediction: predicted disposition based on classification threshold and ExoMiner score
     (string)
uid, target_id, sector_run, tce_plnt_num, tce_period, tce_duration, tce_time0bk, tce_depth, ruwe
    , tce_prad, tce_max_mult_ev, matchedToiId, score_PC,score_std_PC,score_AFP,score_std_AFP,
    score_NTP,score_std_NTP,prediction
27397122-1-S14-86, 27397122, 14-86, 1, 12.732, 3.292, 1689.460, 14560.665, 1.046, 25.507,
    117.426, not_matched, 0.826, 0.240, 0.172, 0.239, 0.002, 0.003, PC
```

Figure 1: Snippet of the `ExoMiner Pipeline` prediction table output (CSV format). Float values have been truncated for display purposes.

the red dashed line shows the standard error of the mean (SEM) envelope for the bins. A larger number of transit observations, as well as cleaner transits, result in a tighter envelope. For a more detailed explanation of the data preprocessing steps we refer the reader to [Val+22; Val+25].

The title of each plot in this page provides the name of time series along with the number of transits phase-folded and binned to create it. The x-axis shows the phase in hours centered in the detected epoch, with the "transit-view" plots showing the phase in $[-2.5, 2.5]$ transit durations while the "full-orbit" plots show the complete orbital period. The y-axis of the flux time series plots is the relative flux, and that for the centroid motion time series is the centroid offset relative to the target's TIC coordinates. In several cases, this offset can be unreliable and may differ from the vector displacements reported in standard SPOC centroid offset plots which rely on difference image analysis. This discrepancy arises because the measured center of light often contains a static baseline offset from the catalog position due to background crowding (blending), stellar proper motion, or slight coordinate systematic biases. The "full-orbit trend" plot shows the phase-folded and binned residual of the Savitzky-Golay filter. As legacy, the momentum dump flag time series is shown as part of this page, although no longer used by ExoMiner as an input feature.

3.2 Weak Secondary

The weak secondary plots shown in Figure 2 are centered on the detected secondary transit. The top plot shows the phase-folded flux time series and the red triangle at the bottom of the plot points to the primary transit midpoint. The bottom plots show the "transit-view" centered also on the secondary transit midpoint and show the phase range in $[-2.5, 2.5]$ transit durations. If the primary transit is within that interval, a red triangle at the bottom of the plot will indicate its location. **Note:** While the SPOC pipeline gaps the primary transits of previous TCEs detected in the same run and for the same target star, the ExoMiner Pipeline plots do not follow this procedure. *Consequently, vetters should be aware that the flux plots may display transits associated with other previously detected TCEs.*

3.3 Odd-even

For the odd-even transit depth comparison, the plot in Figure 3 shows the phase-folded and binned transit-view flux time series for the odd (green) and even (red) transits. The two time series were created by using the same TCE ephemerides but using alternating phases. Similar to the plots in Figure 4, the SEM envelope for both odd and even flux time series is shown using dashed lines. The data points used to create the binned versions are shown in black with some transparency.

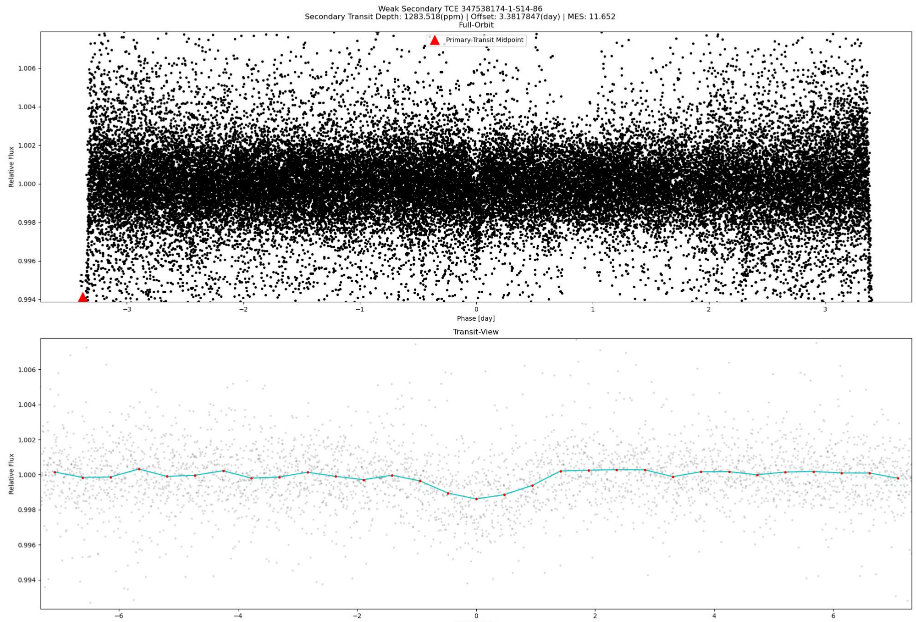

Figure 2: Weak secondary plots for `TIC 347538174.1`.

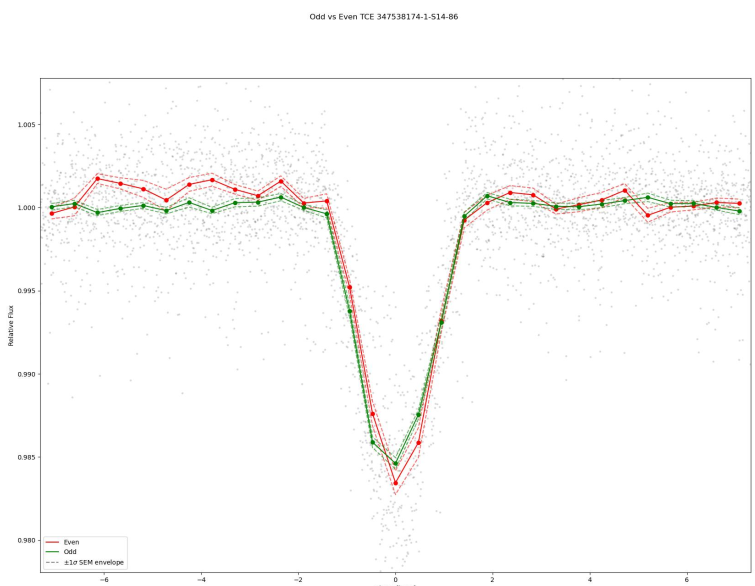

Figure 3: Odd-even plots for `TIC 347538174.1`.

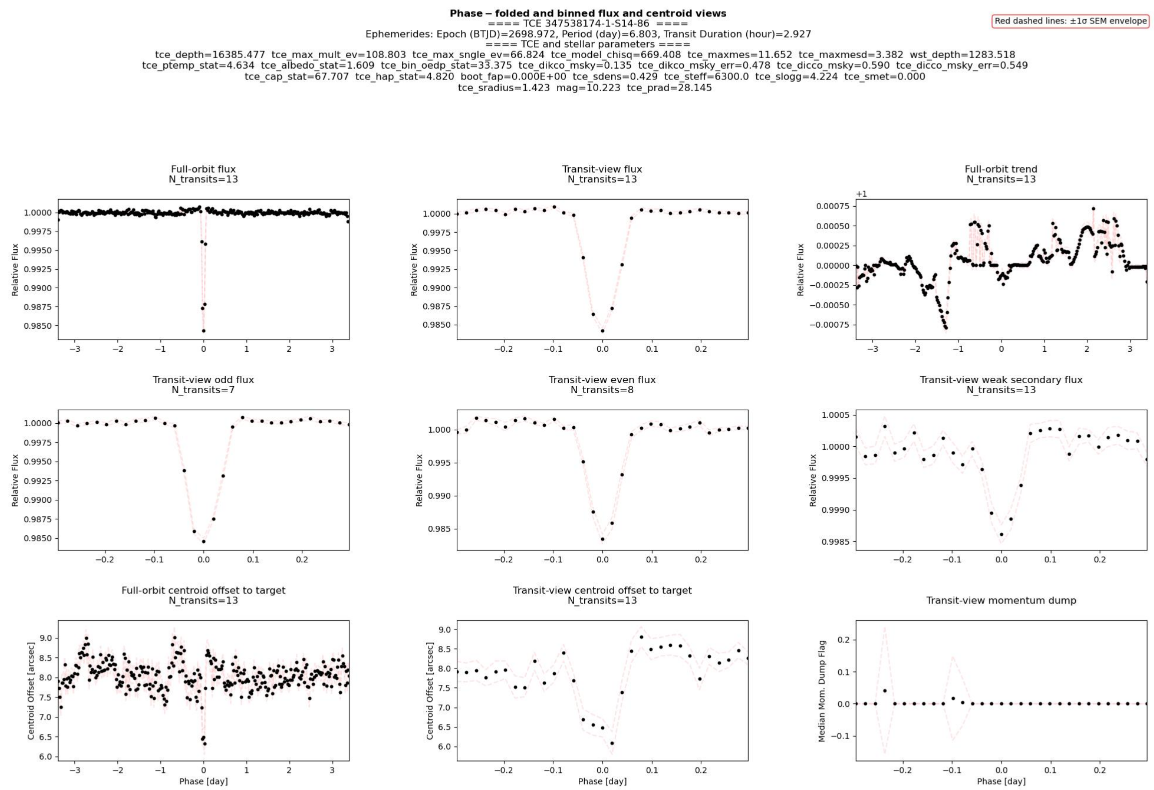

Figure 4: Phase-folded and binned flux and centroid plots for `TIC 347538174.1`.

3.4 Periodogram

The periodogram page shown in Figure 5 shows the periodograms for both the flux time series as well as the transit pulse model (TPM) for the SPOC TCE. The TPM time series shows a simple transit pulse created using the detected ephemerides. The figure's header shows the period and corresponding frequency detected by the SPOC pipeline. The top plot shows the periodogram for both the flux and TPM time series. The first five harmonics of the detected period for the TCE are highlighted using red triangles at the bottom of the plot, and the header shows the max amplitude and its corresponding frequency. Both periodograms are also shown using a smoothed version. The bottom plot shows the same data, but each periodogram was normalized by its maximum amplitude. All periodograms are shown in a log-scale in the period domain.

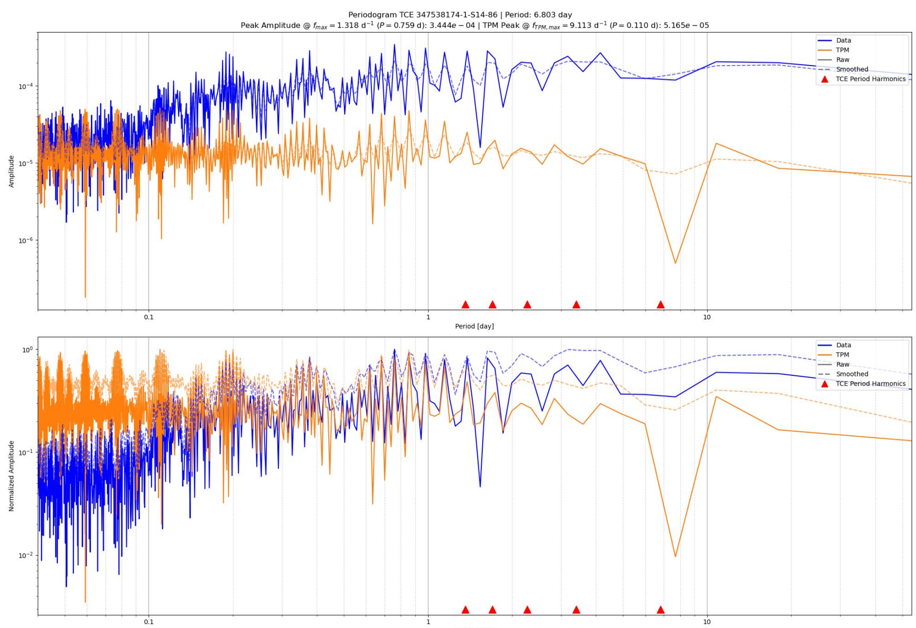

Figure 5: Periodogram plots for `TIC 347538174.1`.

3.5 Difference Image

For each sector in which the target was observed and the SPOC pipeline was able to generate difference images for the TCE, the summary report shows a page with several images related to the difference image data in that sector. Figure 6 shows an example for `TIC 347538174.1` in sector 51. In all these images the location of the target according to its TIC coordinates is shown as a red cross. The title of the figure shows the sector ID, the target's TESS

Magnitude, and the computed quality metric in the SPOC Data Validation (DV) pipeline (correlation between the difference image and the pixel response function centered on the centroid of the difference image). Four different types of images are computed per sector:

- Difference flux: mean image created by subtracting the in-transit image from the out-of-transit image. This image is created in the DV module of the SPOC pipeline.
- Out-of-transit flux: mean image created using all image data for the out-of-transit cadences. This image is created in the DV module of the SPOC pipeline. Any negative pixels are greyed out.
- SNR flux: computed as the difference image flux divided by the uncertainty in each pixel [Twi+18].
- Valid pixels: this is a quality mask that shows which pixels are deemed invalid (e.g., negative out-of-transit image pixels).

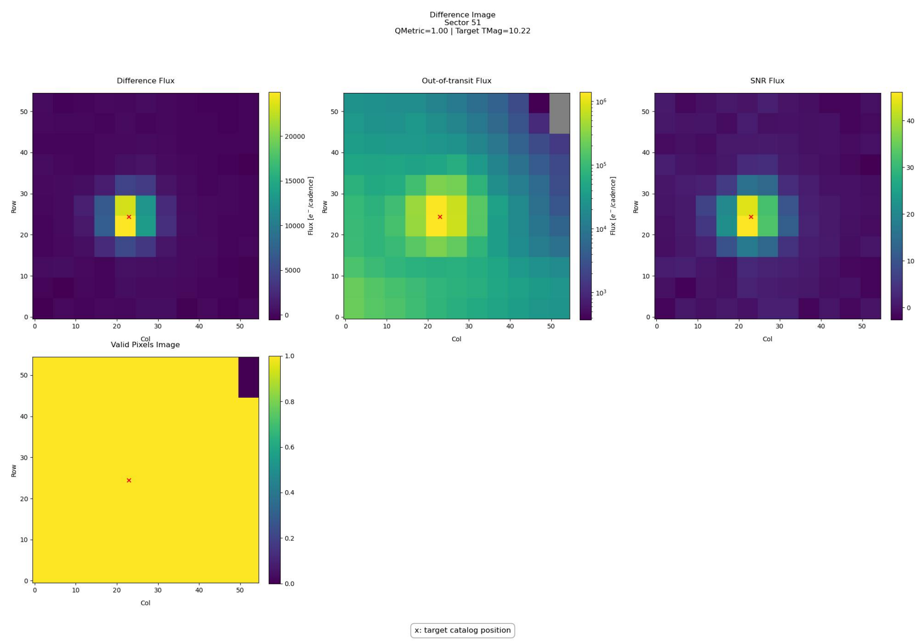

Figure 6: Difference image plots for `TIC 347538174.1` in sector 51.

4 Conclusion

This document provides an overview of the outputs generated by the `ExoMiner Pipeline`. By producing both a detailed tabular prediction file and a visual diagnostic summary report, the pipeline serves as a tool for expert vetters engaged in the vetting and validation of exoplanet candidates from the TESS mission. The prediction table encapsulates crucial metadata, run provenance, and Machine Learning classification scores, while the auxiliary plots give expert vetters insights into the data and the preprocessing steps taken to produce the data ingested by the model, as well as a better understanding of the model behavior and, ultimately, its prediction scores.

Any updates to the `ExoMiner Pipeline` and its outputs will be reflected in future versions of this document.

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